

Unit 5 Review

STAT 4710

November 18, 2025

Roadmap for Unit 5

Lecture 1: Deep learning preliminaries

- Predictive models as graphs
- Training via optimization: gradient descent
- Hands-on: simple NN on MNIST in R torch

Lecture 2: Neural networks

- Multi-layer predictive models
- Stochastic gradient descent
- Hands-on: multi-layer NN for MNIST in R torch with newly learned training options

Lecture 3: Deep learning for images

- Image classification
- Convolutional neural networks
- Hands-on: CNN for MNIST in R torch

Lecture 4: Deep learning for text

- Document classification
- Recurrent neural networks and transformers

Lecture 1: preliminaries

- Softmax activation function (definition)
- Cross-entropy loss (definition)
- Convex functions and gradient descent

Lecture 2: multi-layer neural networks & training

- ReLU activation function (definition)
- Compute the number of parameters in a fully connected layer
- Stochastic gradient descent:
 - mini-batches, batch size, epoch
 - how to compute the number of SGD steps in several epochs
 - Benefits of SGD over GD
- Regularization techniques to address overfitting
 - How to judge when overfitting happens?
 - Explicit regularization: weight decay
 - Implicit regularization: early stopping & dropout

Lecture 3: Convolutional neural networks

- Convolution operation: definitions
- Compute the number of parameters in a convolution layer
- Compute the output size of a convolution layer given input & filter size & numbers
- Max pooling: definition
- Compute the output size of a max-pooling layer given input & filter sizes

Lecture 4: Natural language processing

- Definition of a vocabulary (how many tokens)
- RNN structure: Which sets of parameters? What are they for?
- Compute the total number of parameters given input & hidden state sizes
- Transformer structure: positional embedding to encode word order
- RNN v.s. Transformer: RNN in sequence (slow), Transformer in parallel (fast) for encoding and sequential for decoding